

Topic 5

Mensuration

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

Learning focus

Explain every method before practising. Use the original practice questions to check calculation, reasoning and communication.

Topic map

5.1	Units and compound measures
5.2	Perimeter and area
5.3	Circles, sectors and arcs
5.4	Volumes and surface areas
5.5	Similar solids and practical problems

Core foundations and Extended applications are integrated throughout. Study each example, then attempt the practice paper without the notes.

1. Units and area

Convert lengths before calculating. Area is measured in square units: $1 \text{ m}^2 = 10000 \text{ cm}^2$; volume is in cubic units: $1 \text{ m}^3 = 1000000 \text{ cm}^3$. A triangle has area $\frac{1}{2} \times \text{base} \times \text{perpendicular height}$, and a trapezium has area $\frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$.

Worked example

For a trapezium with parallel sides 7 cm and 11 cm and height 4 cm, area = $(7 + 11) \times \frac{4}{2} = 36 \text{ cm}^2$.

Exam check: The sloping edge of a triangle is not its perpendicular height unless it forms a right angle to the base.

2. Circles, arcs and sectors

The circle circumference is $2\pi r$ and area is πr^2 . A sector with central angle θ takes fraction $\frac{\theta}{360}$ of a full circle, so its arc length is $(\frac{\theta}{360})2\pi r$ and area is $(\frac{\theta}{360})\pi r^2$. Include straight radii when asked for sector perimeter.

Worked example

Radius 6 cm, angle 60° : arc = $(\frac{60}{360})2\pi(6) = 2\pi \text{ cm}$; sector area = $(\frac{60}{360})\pi(6^2) = 6\pi \text{ cm}^2$.

Exam check: Distinguish the curved arc from the full perimeter of a sector.

3. Prisms and cylinders

For a prism, volume = cross-sectional area $\times$ length. For a cylinder, volume = $\pi r^2 h$ and curved surface area = $2\pi r h$. Total closed surface area includes both circular ends; distinguish an open container from a closed one.

Worked example

A cylinder with $r = 3 \text{ cm}$ and $h = 8 \text{ cm}$ has volume $\pi(3^2)(8) = 72\pi \text{ cm}^3$.

Exam check: A diameter of 6 cm means $r = 3 \text{ cm}$; do not use 6 in πr^2 .

4. Cones, spheres and pyramids

A cone has volume $(\frac{1}{3})\pi r^2 h$ and curved surface area $\pi r l$, where l is slant height. A sphere has volume $(\frac{4}{3})\pi r^3$ and surface area $4\pi r^2$. A pyramid has volume $(\frac{1}{3}) \times \text{base area} \times \text{perpendicular height}$. Pythagoras often connects a cone height, radius and slant height.

Worked example

For a cone with $r = 3 \text{ cm}$ and $h = 4 \text{ cm}$, $l = 5 \text{ cm}$, so curved area = $15\pi \text{ cm}^2$ and volume = $12\pi \text{ cm}^3$.

Exam check: Use the perpendicular height for volume and slant height for curved surface area.

5. Composite and similar solids

Break a compound solid into standard parts. Add volumes of joined solids; subtract removed hollows. If lengths are multiplied by k , areas multiply by k^2 and volumes by k^3 . For material cost, use the actual exposed surface, including or excluding joined faces as appropriate.

Worked example

Two similar solids have length ratio 2:5. Their volume ratio is $2^3:5^3 = 8:125$.

Exam check: For a joined solid, an internal contact face is not part of the external surface.

Before the practice paper

Method and accuracy

Write the formula or first operation, show the intermediate step, give units when needed and check by substitution, estimation or a sketch. On non-calculator work, keep exact forms when requested.

Practice paper

Topic 5 · Mensuration

Instructions

Original Cambridge-style practice. Answer all questions. Section A is multiple choice; Section B requires working and final answers. Try Section A without a calculator; use a calculator in Section B only where needed.

Candidate name:	
Class / date:	

Total: 32 marks

Section A · Multiple choice

Choose one answer for each question.

1. 1 m² equals how many cm²? [1]

- A 100
- B 1000
- C 10000
- D 1000000

2. Area of a triangle with base 8 cm and height 5 cm? [1]

- A 13
- B 20
- C 40
- D 80

3. Circumference of a circle with radius 4? [1]

- A 4π
- B 8π
- C 16π
- D 32π

4. Volume of a cylinder with radius 2 and height 5? [1]

- A 10π
- B 20π
- C 25π
- D 40π

5. Area of a 90° sector with radius 4? [1]

- A 2π
- B 4π
- C 8π
- D 16π

6. Volume of a cone with radius 3 and height 6? [1]

- A 6π
- B 18π
- C 27π
- D 54π

7. Surface area of a sphere with radius 2? [1]

- A 8π
- B 12π
- C 16π
- D 32π

8. Length scale factor 4 gives volume factor [1]

- A 4
- B 8
- C 12
- D 64

Section B · Structured questions

9. A sector has radius 8 cm and central angle 90° .

(a) Find its arc length in terms of π . [2]

(b) Find its area in terms of π . [2]

(c) Find its perimeter in terms of π . [2]

[Total: 6]

10. A closed cylinder has radius 3 cm and height 10 cm.

(a) Calculate its volume in terms of π . [2]

(b) Calculate its curved surface area in terms of π . [2]

(c) Calculate its total surface area in terms of π . [2]

[Total: 6]

11. A cone has radius 5 cm and perpendicular height 12 cm.

(a) Find its slant height. [2]

(b) Find its volume in terms of π . [2]

(c) Find its curved surface area in terms of π . [2]

[Total: 6]

12. Two similar cylinders have radii in the ratio 3:5. The smaller has volume 54 cm^3 and surface area 72 cm^2 .

(a) Find their volume ratio. [2]

(b) Find the larger volume.

[2]

(c) Find the larger surface area.

[2]

[Total: 6]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Award method marks when the mathematical setup is correct even if an arithmetic slip follows.
Check each part independently.

Section A · Answers and reasons

1. C — 10000. $1\text{ m} = 100\text{ cm}$; square the factor to obtain 10000.
2. B — 20. Triangle area is $\frac{1}{2} \times \text{base} \times \text{perpendicular height}$.
3. B — 8π . Circumference $= 2\pi r = 8\pi$.
4. B — 20π . Volume $= \pi r^2 h = \pi(2^2)(5) = 20\pi$.
5. B — 4π . Quarter-circle area $= \pi(4^2)/4 = 4\pi$.
6. B — 18π . Cone volume $= \pi r^2 h/3 = \pi(9)(6)/3$.
7. C — 16π . Sphere area $= 4\pi r^2 = 4\pi(4)$.
8. D — 64. Volume factor is the cube of the length factor: 4^3 .

Section B · Worked solutions

9. a) $(90/360) \times 2\pi(8) = 4\pi \text{ cm}$ [2]. b) $(90/360) \times \pi(8^2) = 16\pi \text{ cm}^2$ [2]. c) $4\pi + 8 + 8 = 16 + 4\pi \text{ cm}$ [2].

10. a) $\pi(3^2)(10) = 90\pi \text{ cm}^3$ [2]. b) $2\pi(3)(10) = 60\pi \text{ cm}^2$ [2]. c) $60\pi + 2\pi(3^2) = 78\pi \text{ cm}^2$ [2].

11. a) $l = \sqrt{5^2 + 12^2} = 13 \text{ cm}$ [2]. b) $(1/3)\pi(5^2)(12) = 100\pi \text{ cm}^3$ [2]. c) $\pi(5)(13) = 65\pi \text{ cm}^2$ [2].

12. a) $3^3:5^3 = 27:125$ [2]. b) $54 \times 125/27 = 250 \text{ cm}^3$ [2]. c) $72 \times 25/9 = 200 \text{ cm}^2$ [2].

Review your errors

Mark the first wrong step: setup, algebra, arithmetic, units, graph or reasoning. Redo that part without seeing the solution.

Syllabus reference: Cambridge IGCSE Mathematics 0580, 2025-2027,
www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf. Paper style informed by the June 2026 papers and examiner report provided by the user. All practice questions are original.